

Experiment 7.1

Reconstruction of Signal from Samples (Ideal Interpolation)

Program (MATLAB)

```
% Reconstruction of Signal from Samples
```

```
clc;
```

```
clear;
```

```
close all;
```

```
% Original signal frequency
```

```
fm = 5; % Hz
```

```
% Sampling frequency
```

```
fs = 25; % Hz
```

```
% Continuous-time signal
```

```
t = 0:0.001:1;
```

```
x = sin(2*pi*fm*t);
```

```
% Sampling
```

```
ts = 0:1/fs:1;
```

```
xs = sin(2*pi*fm*ts);
```

```
% Reconstruction using interpolation
```

```
xr = interp1(ts,xs,t,'spline');
```

```
% Plotting
```

```
figure;
```

```
subplot(3,1,1);
```

```
plot(t,x,'LineWidth',2);
```

```
grid on;
```

```
title('Original Signal');
```

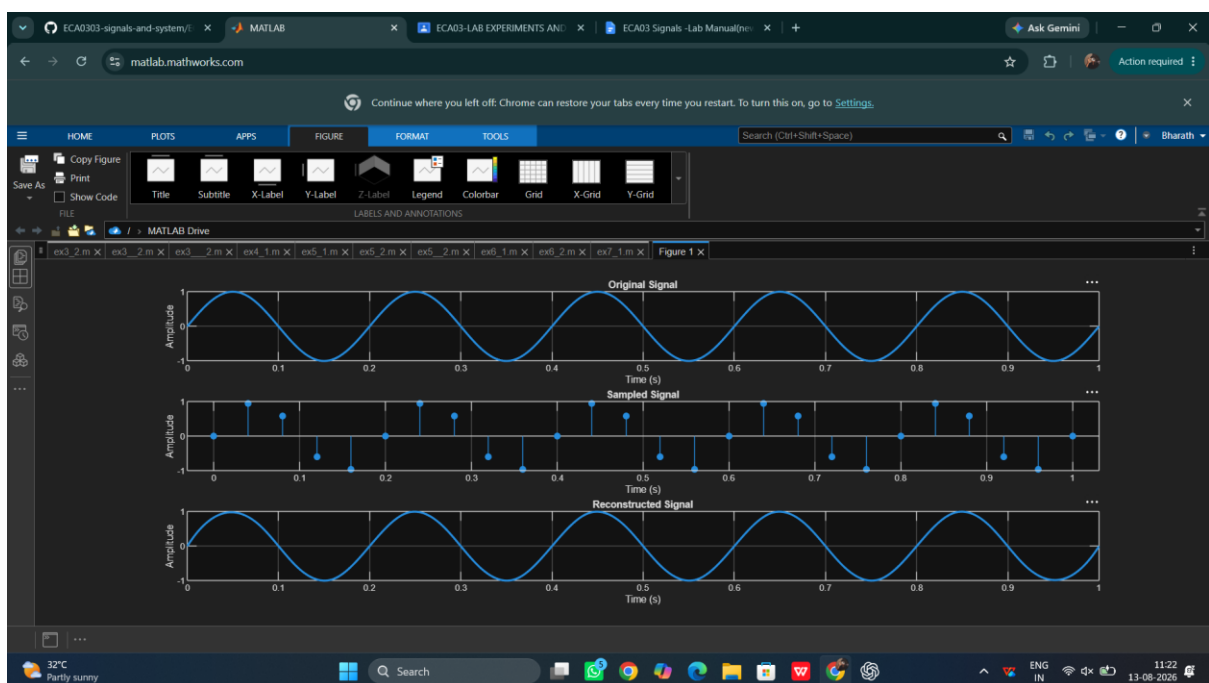
```
xlabel('Time (s)');
```

```

ylabel('Amplitude');
subplot(3,1,2);
stem(ts,xs,'filled');
grid on;
title('Sampled Signal');
xlabel('Time (s)');
ylabel('Amplitude');
subplot(3,1,3);
plot(t,xr,'LineWidth',2);
grid on;
title('Reconstructed Signal');
xlabel('Time (s)');
ylabel('Amplitude');

```

Output Generated Graph



Result

The sinusoidal signal was sampled at **25 Hz**, which is greater than the Nyquist rate of **10 Hz**. Using interpolation, the original signal was reconstructed successfully, demonstrating accurate signal recovery from its samples in accordance with the Sampling Theorem.